

DSCI 522: Individual Assignment 3

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Aim

This project explores the historical population of horses in Canada between 1906 and 1972 for each province.

Data

Horse population data were sourced from the [Government of Canada's Open Data website](#) (Government of Canada, 2017a and Government of Canada, 2017b).

Methods

The Python programming language (Van Rossum and Drake 2009) and the following Python packages were used to perform the analysis: pandas (McKinney 2010), altair (VanderPlas 2018), click (Team 2020), as well as Quarto (Allaire et al. 2022). *Note: this report is adapted from Timbers (2020).*

Results

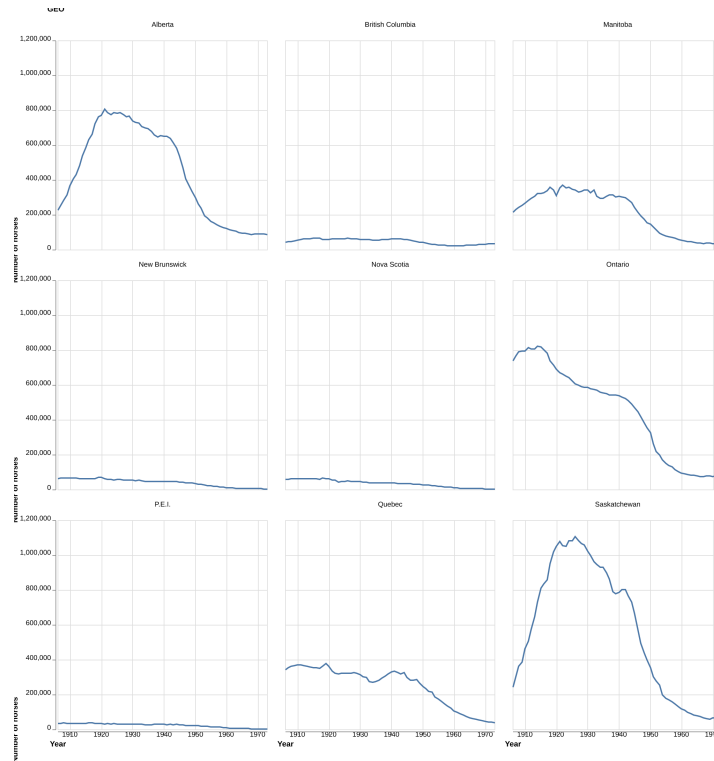


Figure 1: Horse populations for all provinces in Canada from 1906 - 1972.

As shown in Figure 1, Ontario, Saskatchewan, and Alberta have had the highest horse populations. All provinces have had a decline in horse populations since 1940. This is likely due to the rebound of the Canadian automotive industry after the Great Depression and the Second World War. An interesting follow-up visualisation would be car sales per year for each Province over the time period visualised above to further support this hypothesis.

Suppose we were interested in looking in more closely at the province with the highest spread (in terms of standard deviation) of horse populations. We present the standard deviations in Table 1.

Table 1: Standard deviation of historical (1906-1972) horse populations for each Canadian province.

Province	Std
Saskatchewan	377266
Ontario	266435
Alberta	266063

Table 1: Standard deviation of historical (1906-1972) horse populations for each Canadian province.

Province	Std
Manitoba	122404
Quebec	111411
New Brunswick	22019.5
Nova Scotia	19879.3
British Columbia	14945.7
P.E.I.	11355.7

Note that we define standard deviation (of a sample) as

$$s = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}}$$

As shown in Table 1, Saskatchewan had the largest spread in horse population values.

Additionally, note that in Table 1 we consider the sample standard deviation of the number of horses during the same time span as Figure 1.

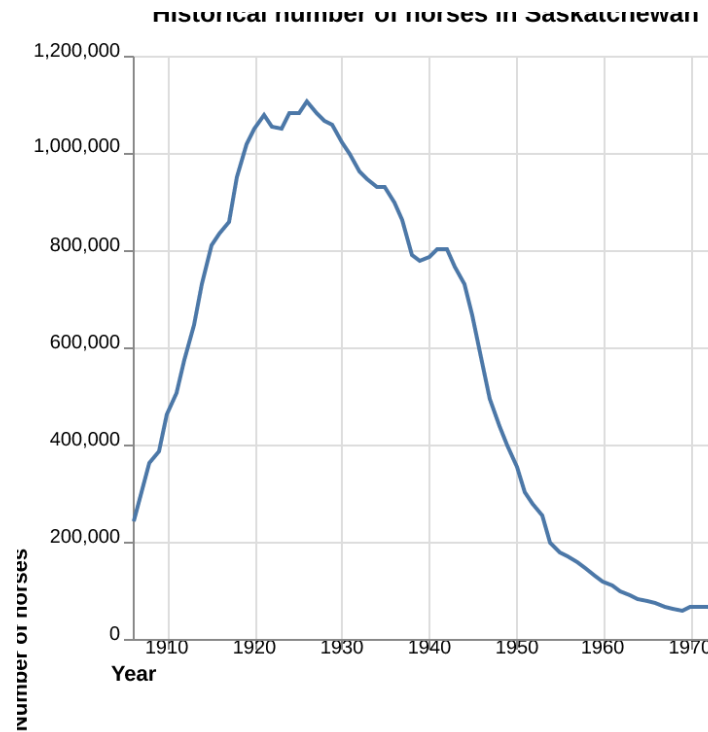


Figure 2: Horse Standard Deviations for all provinces in Saskatchewan from 1906 - 1972.

In Figure 2, we zoom in on Saskatchewan, which had the largest spread of horse population values.

References

Horse population data were sourced from the [Government of Canada's Open Data website](#) (Government of Canada 2017a, 2017b)

Quarto documentation was sourced from [Quarto-cli Github Page](#) (Allaire et al. 2022)

Altair was sourced from [Altair: Interactive Statistical Visualizations for Python](#) (VanderPlas 2018)

Click Documentation was sourced from [Click Documentation Page](#) (Team 2020)

Pandas Documentation (McKinney 2010)

equine_numbers_value_canada_parameters Github Page [equine_numbers_value_canada_parameters Github Page](#) (Timbers 2020)

Knitr Documentation was sourced from [Implementing Reproducible Research](#) (Xie 2014)

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